Cont. Data Structures-I

Session Objectives

- Understand the meaning of data structures and why we use them.
- Common data structures in Python
- Understand what lists are.
- Understand common methods and operations associated with lists.
- Understand the meaning of list comprehension
- Understand what tuples are.
- Understand common methods and operations associated with tuples.
- Understand the Comparison between Lists and Tuples

```python
# Basic List Operations:
# len() -> It returns the count of elements in a list.
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']

print(len(country_list))
```

```
19
```

```python
# min(), max() , sum() -> Continuous Column [Numerical Columns]
num_list = [11,22,33,44,55,66,77,88,99,110]
print(len(num_list)) # 10
print(min(num_list)) # 11
print(max(num_list)) # 110
print(sum(num_list)) # ?
```

```
10
11
110
605
```

```python
# min(), max() , sum() -> Continuous Column [Numerical Columns]
num_list = [11,22,33,44,55,66,77,88,99,110,False,True] # 'mixed Dtype'
# int(False) = 0 & int(True) = 1
print(len(num_list)) # 12
print(min(num_list)) # False - 0
print(max(num_list)) # 110
print(sum(num_list)) # 606
```

```
12
False
110
606
```

```python
print(int(False))
```

```
0
```

```python
print(int(True))
```

```
1
```

```python
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']
print(min(country_list)) # 'America'
print(max(country_list)) # 'Ukraine'
```

```
America
Ukraine
```

```python
country_list = ['India','america','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','india','America','Germany','China']
print(min(country_list)) # 'Australia'
print(max(country_list)) # 'india'
```

```
America
india
```

```python
# List are mutable Container
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']
country_list[-2] = 'Sri-Lanka'
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Sri-Lanka', 'China']
```

```python
country_list[7:10] = 'France' # ['F','r','a','n','c',e]
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'F', 'r', 'a', 'n', 'c',
'e', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Sri-Lanka', 'China']
```

```python
country_list[7:13] = ['France']
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'France', 'Nepal', 'Irel
and', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Sri-Lanka', 'China']
```

```python
# Repeating a content of a list (*) ->
char_list = ['a','b','c','d','e',True,False]
print(char_list * 2)
print(len(char_list * 2))
```

```
['a', 'b', 'c', 'd', 'e', True, False, 'a', 'b', 'c', 'd', 'e', True, False]
14
```

Adding items in a list:

1. `.append()` : Add one item at the end.
2. `.extend()` : Add all items from another list.
3. `.insert()` : Insert an items at specific position.

```python
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']
country_list.append('Singapore')
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore']
```

```python
# .extend() -> connecting the 2 list
new_country_list = ['Poland','UK','Brazil','Isreal','Austria','Japan']
country_list.extend(new_country_list)
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'P
oland', 'UK', 'Brazil', 'Isreal', 'Austria', 'Japan']
```

```python
# .insert() -> (pos,val)
country_list.insert(-5,'Qatar')
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'P
oland', 'Qatar', 'UK', 'Brazil', 'Isreal', 'Austria', 'Japan']
```

```python
# country_list.insert(index, object)
country_list.insert(4,['Turkey','Myanmar','Thailand','Malaysia'])
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', ['Turkey', 'Myanmar', 'Thailand', 'Malaysia'], 'Australia', 'So
uth Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India',
'America', 'Germany', 'China', 'Singapore', 'Poland', 'Qatar', 'UK', 'Brazil', 'Isreal', 'Austria', 'Japan']
```

```python
# Adding a list '+' operators [Concatenation]
weekday_list = ['Mon','Tue','Wed','Thur','Fri']
weekend_list = ['Sat','Sun']
week_list = weekday_list + weekend_list
print(week_list)
print(type(week_list))
```

```
['Mon', 'Tue', 'Wed', 'Thur', 'Fri', 'Sat', 'Sun']
<class 'list'>
```

```python
# Indexing in Nested List:
# 2-Dimension[rows,cols]
nested_list = [
    [11,22,33,44,55],
    ['a','b','c','d','e'],
    [True,True,False,False,True]
]
print(nested_list[0][1]) # 22
print(nested_list[-1][-1]) # True
print(nested_list[2][-3]) # False
print(nested_list[-2][-5]) # 'a'
print(nested_list[2][4]) # True
```

```
22
True
False
a
True
```

```python
# # 2-Dimension[rows,cols]
# Slicing[Start = 0 : Stop = Length(Non-Inclusive) : Step = +1(dir,mag)]
nested_list = [
    [11,22,33,44,55],
    ['a','b','c','d','e'],
    [True,True,False,False,True]
]
print(nested_list[0:2]) # 1st Two rows
```

```
[[11, 22, 33, 44, 55], ['a', 'b', 'c', 'd', 'e']]
```

```python
print(nested_list[::-1]) # reversed List [row-wise] [Bottom_to_Top]
```

```
[[True, True, False, False, True], ['a', 'b', 'c', 'd', 'e'], [11, 22, 33, 44, 55]]
```

```python
nested_list[::-1]
```

```
[[True, True, False, False, True],
 ['a', 'b', 'c', 'd', 'e'],
 [11, 22, 33, 44, 55]]
```

```python
print(nested_list[::-1][0:2])
```

```
[[True, True, False, False, True], ['a', 'b', 'c', 'd', 'e']]
```

```python
print(nested_list[-1][1:4]) # True,False,False
```

```
[True, False, False]
```

```python
# Comparison List:
_list1 = [1,2,3,4,5]
_list2 = [1,2,3]
result = (_list1 == _list2)
print(result) # False
```

```
False
```

```python
# Comparison List:
_list1 = [1,2,3,4,5]
_list2 = [1,2,3,4,5]
result = (_list1 == _list2)
print(result) # True
```

```
True
```

```python
# Comparison List:
_list1 = [1,2,3,4,5,6,7]
_list2 = [1,2,3,4,5]
result = (_list1 != _list2)
print(result) # True
```

```
True
```

```python
# Comparison List:
_list1 = [1,2,3,4,5,6,7]
_list2 = [1,2,3,4,5]
result = (_list1 >= _list2)
print(result) # True
```

```
True
```

```python
# Comparison List:
_list1 = [1,2,3,4,5,6,7]
_list2 = [1,2,3,4,5]
result = (_list1 <= _list2)
print(result) # False
```

```
False
```

```python
# Comparison List:
_list1 = [1,2,3,4,5,6,7]
_list2 = ['1','2','3','4','5']
result = (_list1 != _list2)
print(result) # True
```

```
True
```

```
# Comparison List:
_list1 = [1,2,3,4,5,6,7]
_list2 = ['1','2','3','4','5']
result = (_list1 <= _list2)
print(result) # Error
TypeError: '<=' not supported between instances of 'int' and 'str'
```

```python
# Comparison List:
_list1 = ['a','b','c']
_list2 = ['1','2','3','4','5']
result = (_list1 >= _list2)
print(result) # True
```

```
True
```

Removing Items From a List:

1. `.remove()` : Remove the First Occurence of an element.
2. `.pop()` : Removes by index, returns value can be stored in a new variable
3. `del` -> Delete by Index or Slicing
4. `.clear()` -> Clear the List.

```python
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', ['Turkey', 'Myanmar', 'Thailand', 'Malaysia'], 'Australia', 'So
uth Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India',
'America', 'Germany', 'China', 'Singapore', 'Poland', 'Qatar', 'UK', 'Brazil', 'Isreal', 'Austria', 'Japan']
```

```python
country_list.remove(['Turkey', 'Myanmar', 'Thailand', 'Malaysia'])
```

```python
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'P
oland', 'Qatar', 'UK', 'Brazil', 'Isreal', 'Austria', 'Japan']
```

```
country_list.remove(['Poland', 'Qatar', 'UK', 'Brazil', 'Isreal', 'Austria', 'Japan'])
ValueError: list.remove(x): x not in list
```

```python
country_list.remove('Poland')
print(country_list)
```

```
['India', 'America', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germa
ny', 'Nepal', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'Q
atar', 'UK', 'Brazil', 'Isreal', 'Austria', 'Japan']
```

```python
# .remove() -> Remove first occurrence of value.
country_list.remove('America')
print(country_list)
```

```
['India', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepa
l', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'Qatar', 'U
K', 'Brazil', 'Isreal', 'Austria', 'Japan']
```

```python
# .pop(idx) -> Default idx = -1 , and stored the popped element in a new variable
pop_country = country_list.pop(-2) # 'Austria'
print(pop_country)
print(country_list)
```

```
Austria
['India', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepa
l', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'Qatar', 'U
K', 'Brazil', 'Isreal', 'Japan']
```

```python
pop_country = country_list.pop() # 'Japan'
print(pop_country)
print(country_list)
```

```
Japan
['India', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepa
l', 'Ireland', 'Iran', 'Pakistan', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'Qatar', 'U
K', 'Brazil', 'Isreal']
```

```python
# del with index
del country_list[12] # 'Pakistan'
print(country_list)
```

```
['India', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepa
l', 'Ireland', 'Iran', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'Qatar', 'UK', 'Brazil',
'Isreal']
```

```python
# del with slicing
del country_list[-4:] # ['Qatar', 'UK', 'Brazil', 'Isreal']
print(country_list)
```

```
['India', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepa
l', 'Ireland', 'Iran', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore']
```

```python
print(pop_country)
country_list.append(pop_country)
print(country_list)
```

```
Japan
['India', 'Ukraine', 'New Zealand', 'Australia', 'South Africa', 'Russia', 'Italy', 'Spain', 'Germany', 'Nepa
l', 'Ireland', 'Iran', 'Canada', 'India', 'America', 'Germany', 'China', 'Singapore', 'Japan']
```

```python
# .clear()-> It removes every elements of a list.
country_list.clear()
print(country_list) # []
```

```
[]
```

```python
# del with list without indexing or slicing
del country_list # Delete your complete list
```

```python
print(country_list) # NameError
NameError: name 'country_list' is not defined
```

```python
# .Sorting [Ascending | Descending]
# Continuous Column [Low to High and vice-versa]
# Categorical Column [Alphabetical Sorting] [ASCII]
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']

country_list.sort() # default : Ascending
```

```python
print(country_list)
```

```
['America', 'America', 'Australia', 'Canada', 'China', 'Germany', 'Germany', 'India', 'India', 'Iran', 'Irelan
d', 'Italy', 'Nepal', 'New Zealand', 'Pakistan', 'Russia', 'South Africa', 'Spain', 'Ukraine']
```

```python
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']

country_list.sort() # A-Z
print(country_list[::-1]) # Reverse but with logic
```

```
['Ukraine', 'Spain', 'South Africa', 'Russia', 'Pakistan', 'New Zealand', 'Nepal', 'Italy', 'Ireland', 'Iran',
'India', 'India', 'Germany', 'Germany', 'China', 'Canada', 'Australia', 'America', 'America']
```

```python
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','India','America','Germany','China']

country_list.sort(reverse = True) # Z-A
print(country_list)
```

```
['Ukraine', 'Spain', 'South Africa', 'Russia', 'Pakistan', 'New Zealand', 'Nepal', 'Italy', 'Ireland', 'Iran',
'India', 'India', 'Germany', 'Germany', 'China', 'Canada', 'Australia', 'America', 'America']
```

```python
# ASCII 'a' > 'A'
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','india','america','germany','china']
country_list.sort() # A-Z & a-z
print(country_list)
```

```
['America', 'Australia', 'Canada', 'Germany', 'India', 'Iran', 'Ireland', 'Italy', 'Nepal', 'New Zealand', 'Pa
kistan', 'Russia', 'South Africa', 'Spain', 'Ukraine', 'america', 'china', 'germany', 'india']
```

```python
# ASCII 'a' > 'A'
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','india','america','germany','china']
country_list.sort(key = str.lower) # every elements are first converted into lowercase and then sorted
print(country_list)
```

```
['America', 'america', 'Australia', 'Canada', 'china', 'Germany', 'germany', 'India', 'india', 'Iran', 'Irelan
d', 'Italy', 'Nepal', 'New Zealand', 'Pakistan', 'Russia', 'South Africa', 'Spain', 'Ukraine']
```

```python
# ASCII 'a' > 'A'
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','india','america','germany','china']
country_list.sort(key = str.lower , reverse = True)
print(country_list)
```

```
['Ukraine', 'Spain', 'South Africa', 'Russia', 'Pakistan', 'New Zealand', 'Nepal', 'Italy', 'Ireland', 'Iran',
'India', 'india', 'Germany', 'germany', 'china', 'Canada', 'Australia', 'America', 'america']
```

```python
# ASCII 'a' > 'A'
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','india','america','germany','china']
country_list.sort(key = str.upper) # every elements are first converted into uppercase and then sorted
print(country_list)
```

```
['America', 'america', 'Australia', 'Canada', 'china', 'Germany', 'germany', 'India', 'india', 'Iran', 'Irelan
d', 'Italy', 'Nepal', 'New Zealand', 'Pakistan', 'Russia', 'South Africa', 'Spain', 'Ukraine']
```

```python
country_list = ['India','America','Ukraine','New Zealand','Australia','South Africa',
                'Russia','Italy','Spain','Germany','Nepal','Ireland','Iran','Pakistan',
                'Canada','india','america','germany','china']
country_list.reverse()
print(country_list)
```

```
['china', 'germany', 'america', 'india', 'Canada', 'Pakistan', 'Iran', 'Ireland', 'Nepal', 'Germany', 'Spain',
'Italy', 'Russia', 'South Africa', 'Australia', 'New Zealand', 'Ukraine', 'America', 'India']
```

```
Doubt Session:
def sum(val1, val2):
    return val1 + val2;

def sum(val1, val2, val3):
    return val1 + val2 + val3;

def sum(val1, val2, val3, val4):
    return val1 + val2 + val3 + val4;

def add(*val):
    return sum(val)
```

```python
# tuple = immutable container
_tuple = (1,2,3,4,5,['a','b','c','d','e'])
print(_tuple)
print(type(_tuple)) # 'tuple'
```

```
(1, 2, 3, 4, 5, ['a', 'b', 'c', 'd', 'e'])
<class 'tuple'>
```

```
_tuple[-1]
```

```
['a', 'b', 'c', 'd', 'e']
```

```
type(_tuple[-1])
```

```
list
```

```
_tuple[-1].append('f')
```

```
print(_tuple)
```

```
(1, 2, 3, 4, 5, ['a', 'b', 'c', 'd', 'e', 'f'])
```